ENTRY 034 The Batch-Circuit Dataset — August 16, 2026 — Status: COMPLETE

Entry 034 — From 10 Hand-Picked Circuits to 116 Sampled Ones

Batch Methodology Surfaces a New Failure Pattern the Reference Sets Never Showed

1. Why Batch Instead of One-Off

Every finding from Entry 022 through Entry 033 rested on 5 hand-picked circuits per chip (adjacent / near / mid / far / one local GHZ) -- the right tool for chasing an individual bug, but a 10-point sample for any claim about how the model behaves in general. entry034_batch_dataset.py instead samples qubit pairs stratified across six BFS hop-distance bins (1, 2-3, 4-7, 8-15, 16-25, 26+), builds a plain Bell circuit for each, runs it through the same Aer/MPS ground-truth pipeline used since Entry 022 (4 seeds × 4096 shots, optimization_level=3), and records the v4.1 (Entry 032) prediction alongside it -- plus a smaller batch of 3-qubit GHZ circuits for structural variety. Every result is checkpointed to JSON as it completes, so the batch can run in resumable chunks.

chip	bell circuits	ghz circuits	total	mean gap	median gap	worst gap
Kyiv	45	10	55	5.38 pts	2.18 pts	31.39 pts
Sherbrooke	51	10	61	7.48 pts	2.14 pts	40.89 pts

116 circuits total, roughly 12x the combined size of every reference set built so far this project.

2. The Expected Pattern: Gap Grows With Route Length

Bucketing by hop distance confirms what the hand-picked circuits suggested but never proved at scale: adjacent pairs (1 hop) predict almost perfectly (median gap 0.1-0.2 points on both chips), and the median gap climbs steadily to 4-14 points by the 16-25 hop bin. Correlation between gap and hop distance across all 96 Bell circuits is +0.20 -- real but modest, because hop distance alone does not explain the worst outliers (below).

3. The New Finding: Short Routes That Collapse to the Floor

Sorting by gap regardless of distance surfaces something the 5-circuit reference sets never could: 7 circuits, 5 of them on Sherbrooke, whose REAL Aer result sits within 2 points of the 50% floor -- the outcome of complete entanglement failure -- despite short 3-9 hop routes the model predicts should succeed 60-91% of the time.

chip	pair	hops	Aer (real)	v4.1 predicts	gap
Sherbrooke	3 → 6	3	50.0%	88.5%	38.5 pts
Sherbrooke	2 → 6	4	50.2%	91.1%	40.9 pts
Sherbrooke	11 → 22	9	48.9%	77.2%	28.3 pts
Sherbrooke	28 → 5	7	50.3%	84.8%	34.5 pts
Kyiv	95 → 72	8	50.0%	60.9%	10.9 pts

Physical qubit 6 (readout error 25.7%, roughly 4-8x every neighboring qubit's readout error on Sherbrooke) and qubit 5 (T2 = 78.9us against a 172.7us T1 -- the largest T1/T2 split on any route in this batch) both recur across these routes. Neither value alone is extreme enough in the current model to explain a full collapse to the floor -- the model's readout and decoherence terms would only cost these circuits 10-20 points, not 35-40.

Entry 034 — Conclusion

The batch approach does exactly what it was proposed to do: instead of finding one bug at a time by hand-picking a circuit, running many circuits and aggregating surfaced a failure mode -- short routes that still crash to the 50% floor -- with 7 independent examples on the first pass, concentrated on two specific physical qubits. That is a concrete, high-confidence lead for the next entry, discovered by frequency rather than by guessing which circuit to isolate next.

4. What This Dataset Is Also For

Beyond bug-hunting, every one of the 116 records is saved with its chip, qubit pair, BFS hop distance, real transpiled edge/gate counts, worst raw edge error and worst T1 along the BFS path, the Aer ground truth, the v4.1 prediction, and the gap -- circuit-and-calibration features paired with a measured label. This is the first dataset in the project large enough to be a plausible starting point for the ML/GNN phase discussed since Entry 033, and it will keep growing the same way as more circuits get added in resumable batches.

5. Next Step

Entry 035: isolate the qubit-5/qubit-6 (Sherbrooke) floor-collapse outliers against Aer the same way Entries 028, 029, and 033 isolated earlier anomalies -- gate error, decoherence, and readout separately -- to find what specifically drives a short route all the way to 50%, since none of the three known channels individually accounts for a drop this size.

Research Log — Index Addendum (Entry 034)

Entry	Title	Date	Status
Entry 034	The Batch-Circuit Dataset (116 circuits, Kyiv + Sherbrooke)	Aug 16, 2026	Complete

Script: entry034_batch_dataset.py (resumable, checkpointed). Data: quantumbridge_data/entry034_batch_dataset.json (116 records), quantumbridge_data/entry034_batch_analysis.json (aggregated bins and outlier list). Surfaces a new floor-collapse pattern on 7 short routes for Entry 035 to isolate. Also the first dataset scaled for the ML/GNN phase.